INCH-POUND
MIL-DTL-7793/2B
19 October 2012
SUPERSEDING
MIL-M-7793/2A
15 September 1988

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL, 28 VDC, ROUND

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be
used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

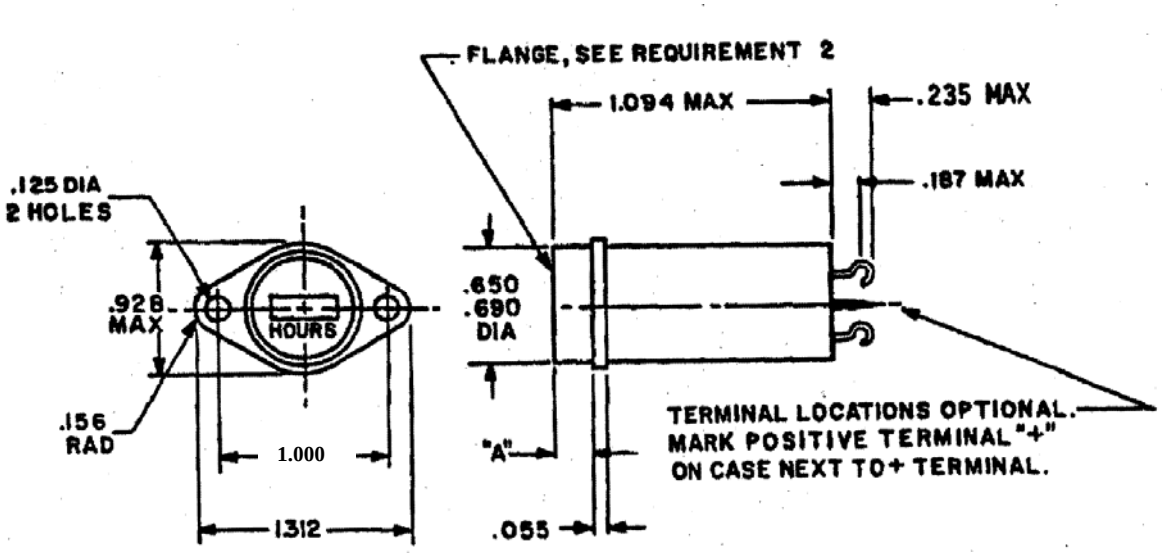


FIGURE 1. Meter configuration.

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REQUIREMENTS:

1. Dimensions. Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

2. Mounting flange. Meters with fixed flanges shall be tin plated on the front surface of the mounting flange in accordance with ASTM-B545 to provide an electrical ground for the case. In accordance with Table I, meters may be furnished with no flange or with adjustable split mount flanges in accordance with MS3345-1.

3. Life accuracy. Meter accuracy shall be ±0.1 percent from -85 to 257° F (-65 to 125° C) and 10 to 34 VDC.

4. Power consumption. The average power consumption at 28 volts direct current (VDC) shall not be greater than 85 milliwatts.

5. Ripple voltage. The meter shall continue to operate as specified in MIL-DTL-7793 when subjected to a cyclic peak of ripple voltage (see Note 3) of less than 2.0 VDC and the frequency-voltage coordinates of figure 2.

5.1 Ripple voltage measurement. The ripple voltage shall be measured with a peak reading voltmeter in series with a 4.0-microfarad capacitor. The higher of the two values measured when the voltmeter is successively connected to each of two polarities shall be considered the ripple voltage.

5.2 Frequency characteristics. The frequency components of the ripple voltage shall be within the limits on figure 2 when measured as conducted interference.

6. Operating temperature range. The operating temperature range shall be from -85 to 257° F (-65 to 125° C).

7. Voltage range. The operating voltage shall be from 10 to 34 VDC.

8. Time range. Total readout of the meters shall be 9,999 hours.

9. Transient protection. No temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown on figure 3 and figure 4 (80V and 600V transient, respectively).

9.1 Transient voltage. Transient surge voltages shall be within the limits on figure 3 when converted to their equivalent step functions. Transient spike voltages shall be within the limits shown on figure 4. They shall be not greater than ±600V nor depart from the normal wave form for more than 50 microseconds.

10. Weight. Weight shall be not greater than 0.6 ounce with flange.

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11. Dielectric strength test voltages. The dielectric test voltages shall be 350 V root mean square (rms) at altitude specified in MIL-DTL-7793 and 600 V rms at room ambient pressure and temperature (approximately 25º C).

12. Qualification tests. Qualification tests shall be conducted in accordance with MIL-DTL-7793, except that the accuracy, voltage, temperature, dielectric strength test voltages and accuracy shall be as specified herein.

TABLE I. Dimension A of flange in figure 1.

Military Part Number	Flange Data Dim A ±.015	Military Part Number	Flange Data Dim A ±.015
M7793/2-001	Adjustable per MS3345-1	M7793/2-015	.375
M7793/2-002	No Flange	M7793/2-016	.406
M7793/2-003	.000 (Flush)	M7793/2-017	.438
M7793/2-004	.031	M7793/2-018	.469
M7793/2-005	.062	M7793/2-019	.500
M7793/2-006	.094	M7793/2-020	.531
M7793/2-007	.125	M7793/2-021	.562
M7793/2-008	.156	M7793/2-022	.594
M7793/2-009	.188	M7793/2-023	.625
M7793/2-010	.218	M7793/2-024	.656
M7793/2-011	.250	M7793/2-025	.688
M7793/2-012	.281	M7793/2-026	.719
M7793/2-013	.312	M7793/2-027	.750
M7793/2-014	.344	M7793/2-028	.781

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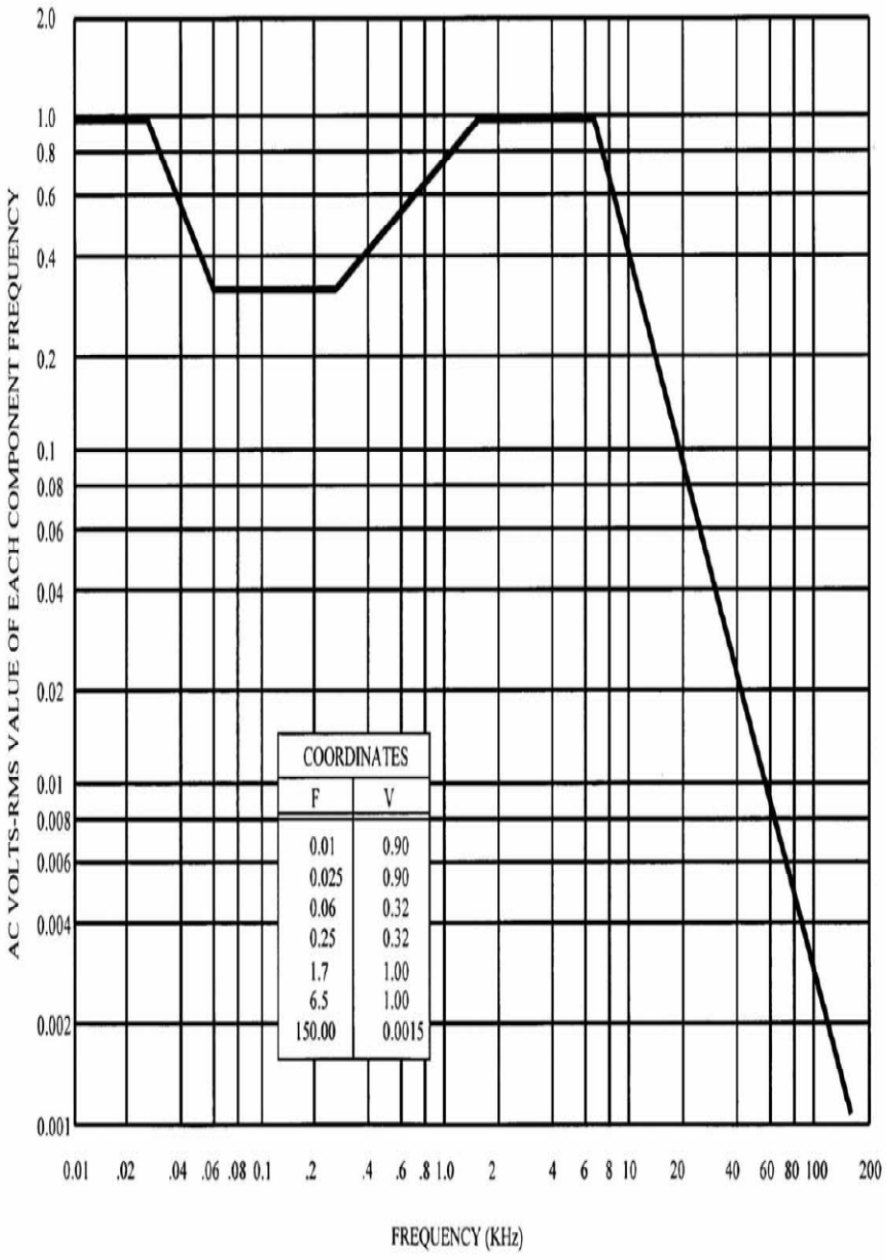


FIGURE 2. Frequency characteristics of ripple in 28 VDC electric systems.

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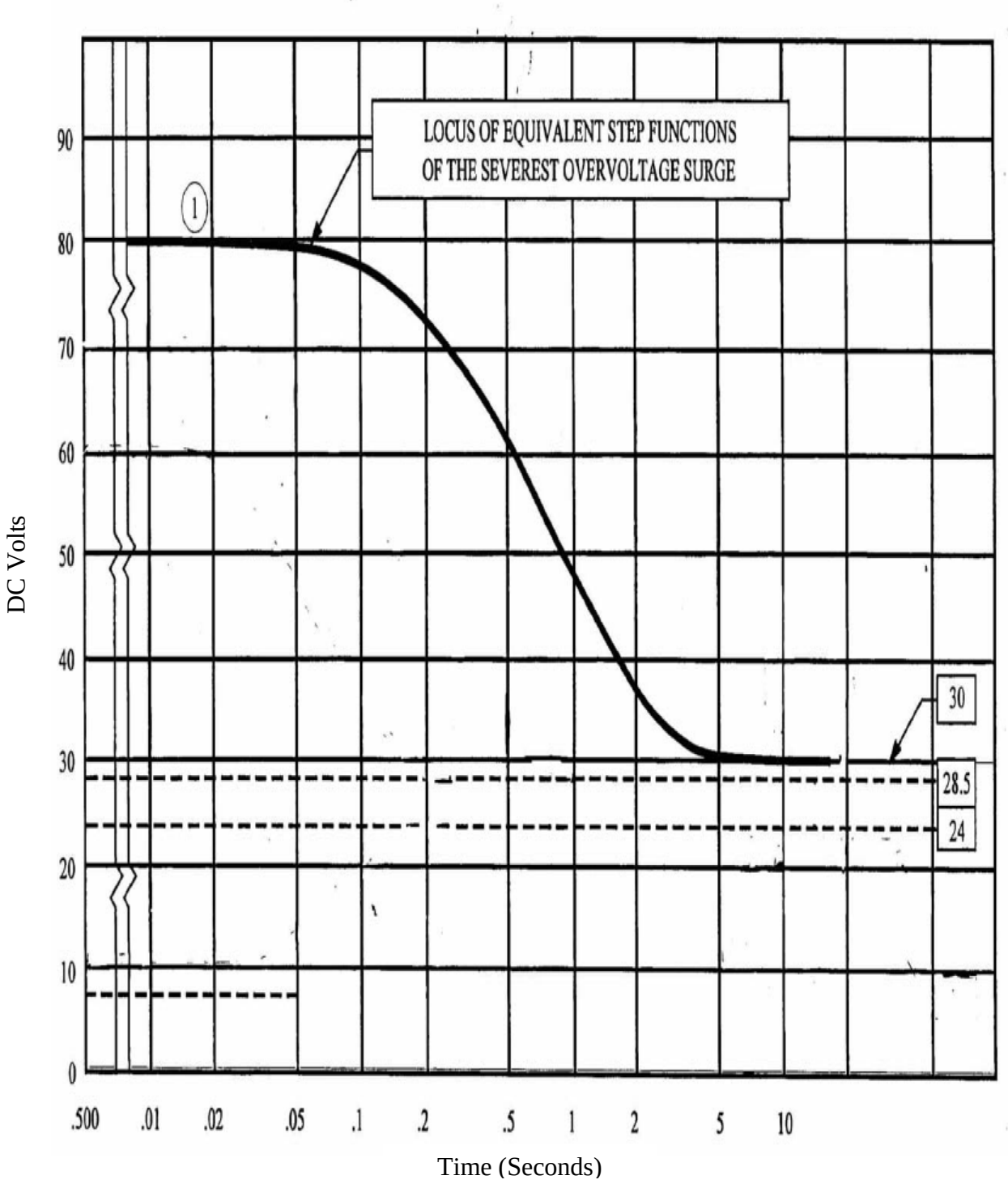


FIGURE 3. Transient surge step function loci limits.

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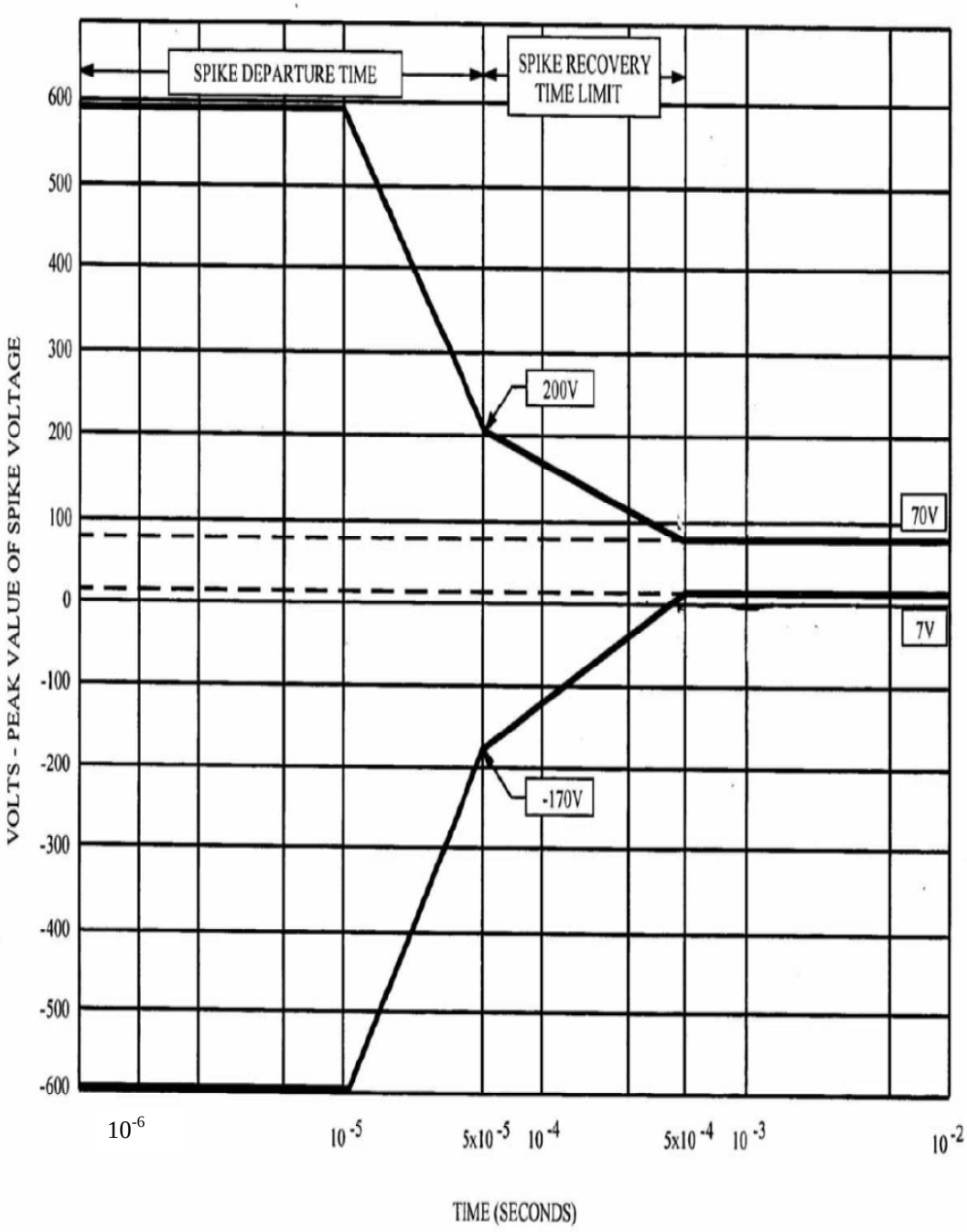


FIGURE 4. Envelope of spike voltages for direct current.

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13. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-7793, this specification sheet references MS3345 and ASTM-B545.

Custodians:
Army - AR
Navy -AS
Air Force - 99

Preparing activity:
Navy - AS
(Project 6645-2012-004)

Review activity:
DLA - GS1

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